

Conditional Eventual Directional Support

Contractive Rayleigh Recurrences and an Explicit Liminf Floor

Prime Dynamics Theory Program

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Abstract

We close the RH-120–RH-127 directional algebra conditionally. Let $x_n = \gamma_n^2$ and suppose eventually $x_{n+1} \leq \rho x_n + q$ with $0 \leq \rho < 1$. Then $\limsup x_n \leq x_* = q/(1 - \rho)$. If $x_* < 1$ and

$$\liminf_n \frac{V_n}{L_n^4 C_n} \geq A_* > 0,$$

the directional candidates satisfy

$$\liminf_n B_n \geq (1 - \sqrt{x_*})^4 A_*.$$

Every threshold strictly below this value is eventually supported. The constant-coefficient floor is sharp. A 4,096-recurrence audit has zero failures. The theorem isolates, but does not prove, the remaining physical all-level packet.

1 Affine limsup theorem

Theorem 1 (Contractive Rayleigh recurrence). *Suppose nonnegative x_n obeys, for all sufficiently large n , $x_{n+1} \leq \rho x_n + q$ with $0 \leq \rho < 1$. Then*

$$\limsup_{n \rightarrow \infty} x_n \leq \frac{q}{1 - \rho}.$$

The same conclusion holds if eventually $\rho_n \leq \rho$ and $q_n \leq q$.

Proof. Iteration gives $x_{N+k} \leq \rho^k x_N + q(1 - \rho^k)/(1 - \rho)$. Let $k \rightarrow \infty$. The nonstationary version is dominated by the same recurrence. $\square$

2 Eventual support closure

Let $A_n = V_n/(L_n^4 C_n)$ and $B_n = (1 - \sqrt{x_n})_+^4 A_n$. RH-125 proves that B_n is a valid lower candidate whenever its finite matrix hypotheses and RH-127 guards hold.

Theorem 2 (Conditional eventual support). *Under Theorem 1, suppose $x_* = q/(1 - \rho) < 1$ and $\liminf A_n \geq A_* > 0$. Then*

$$\liminf B_n \geq (1 - \sqrt{x_*})^4 A_*.$$

Consequently every $\tau < (1 - \sqrt{x_})^4 A_*$ satisfies $B_n > \tau$ for all sufficiently large n .*

Proof. The map $x \mapsto (1 - \sqrt{x})_+^4$ is continuous and decreasing. Combine the limsup bound for x_n with the liminf bound for A_n , then use the definition of liminf. Strict inequality for τ gives eventual support. $\square$

For constant equality $x_{n+1} = \rho x_n + q$ and $A_n = A_*$, the limit equals the displayed floor, proving sharpness. The argument is elementary but its inputs preserve the one-sided matrix logic developed in the previous layers [Horn and Johnson, 1991, Bhatia, 2007].

3 Audit and exact remaining packet

The audit samples 4,096 contractive coefficient caps, subunit fixed points, positive bases, and nonstationary recurrences dominated by those caps. There are zero affine-envelope or support-floor failures. A constant coefficient record approaches the theorem floor to relative error below 10^{-10} .

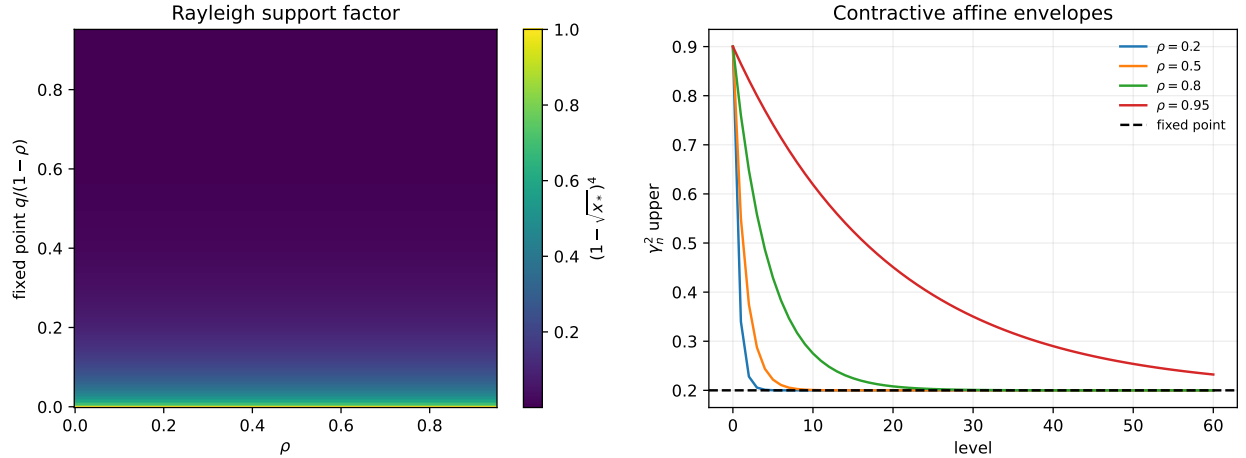


Figure 1: The support factor over the contractive phase diagram and representative affine envelopes.

The remaining physical packet is now explicit: validated all-level coefficients with $\rho < 1$ and $q/(1 - \rho) < 1$, a positive liminf for $V_n/(L_n^4 C_n)$, and either a common operator assembly or RH-127 outward radii. The four finite RH-125 edges cannot establish these eventual facts. No uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.